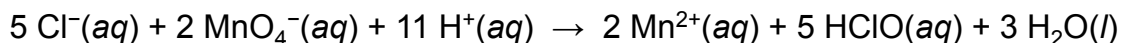


A reaction between chloride and permanganate ions occurs in an acidic solution, as shown below:



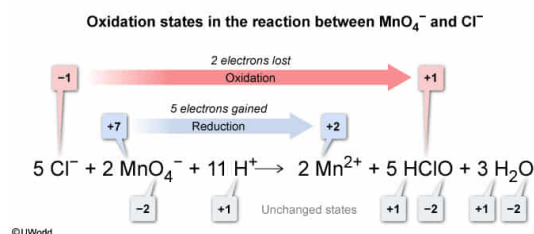
Which element undergoes a change in oxidation state corresponding to a loss of two electrons?

- ☐ A. Mn [35%]
☐ B. O [10%]
✓ ☒ C. Cl [47%]
☐ D. H [5%]

Correct

48% answered correctly

Explanation:



In a **redox reaction**, **oxidation and reduction** always occur together. An atom is reduced by gaining one or more electrons from another atom that loses the electrons (gets oxidized). As a means of **tracking electron transfers** between atoms during reactions, an **oxidation number** (oxidation state) is **assessed** for each atom in a compound relative to the number of electrons in a neutral atom of that element. Atoms with consistent oxidation states (those that rarely vary) are used to determine oxidation states for atoms with variable states (such as metals).

Comparing the oxidation number of an element in the reactants with the oxidation number of that same element in the products indicates whether the element has gained or lost electrons. An increased oxidation state corresponds to a loss of electrons (oxidation), whereas a decreased oxidation state results from gaining electrons (reduction). A change of **one unit** occurs **for each electron** transferred. Elements with unchanged oxidation numbers are not involved in the redox processes.

For the reaction equation given in this question, the assessed oxidation numbers show that the oxidation state of chlorine increases from -1 to $+1$, demonstrating that chlorine loses two negatively charged electrons and is oxidized.

(Choice A) The oxidation state of manganese decreases from $+7$ to $+2$. This means that manganese *gains five* negatively charged electrons and is reduced.

(Choices B and D) The oxidation states of oxygen (-2) and hydrogen ($+1$) remain unchanged during the reaction. No electrons are gained or lost by these elements.

Educational objective:

Oxidation numbers track the transfer of electrons between atoms during a redox reaction. An increased oxidation number corresponds to a loss of electrons (oxidation), whereas a decreased oxidation number results from gaining electrons (reduction). A change of one unit occurs for each electron transferred.

Time spent:0 sec**Subject:**
General Chemistry**QID:**402742**System:**
Interactions of Chemical Substances